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Discovery of Kestrel-7: a Faint, Reddened Globular Cluster Behind the Galactic Bulge

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Abstract. We report the discovery of Kestrel-7, a compact stellar overdensity identified in wide-field survey imaging. Its color–magnitude diagram shows a well-populated red giant branch and a horizontal branch extending blueward of the instability strip, consistent with an old, metal-poor globular cluster. From the horizontal-branch level we estimate a preliminary distance of 11 ± 2 kpc and a foreground reddening of E(B–V) ≈ 0.1.

1. Discovery

Kestrel-7 was flagged by an automated overdensity search at (α, δ) = (274.615°, −24.871°). Proper motions of candidate members are coherent and distinct from the surrounding bulge field.

2. Photometric Properties

The half-light radius is 1.1 arcmin. Several stars on the horizontal branch show photometric variability between survey epochs, suggesting a population of RR Lyrae stars worth dedicated time-series follow-up.

3. Outlook

A time-series campaign in V and I would allow an RR Lyrae–based distance with a precision several times better than the horizontal-branch estimate given here.

Table 1. Summary of key values.

Parameter	Value
RA (J2000)	274.615°
Dec (J2000)	−24.871°
r_h	1.1′
E(B–V)	~0.10
Distance	11 ± 2 kpc

References

Moreau, J., et al. 2022, Journal of Synthetic Astrophysics, 11, 57